ECE 447

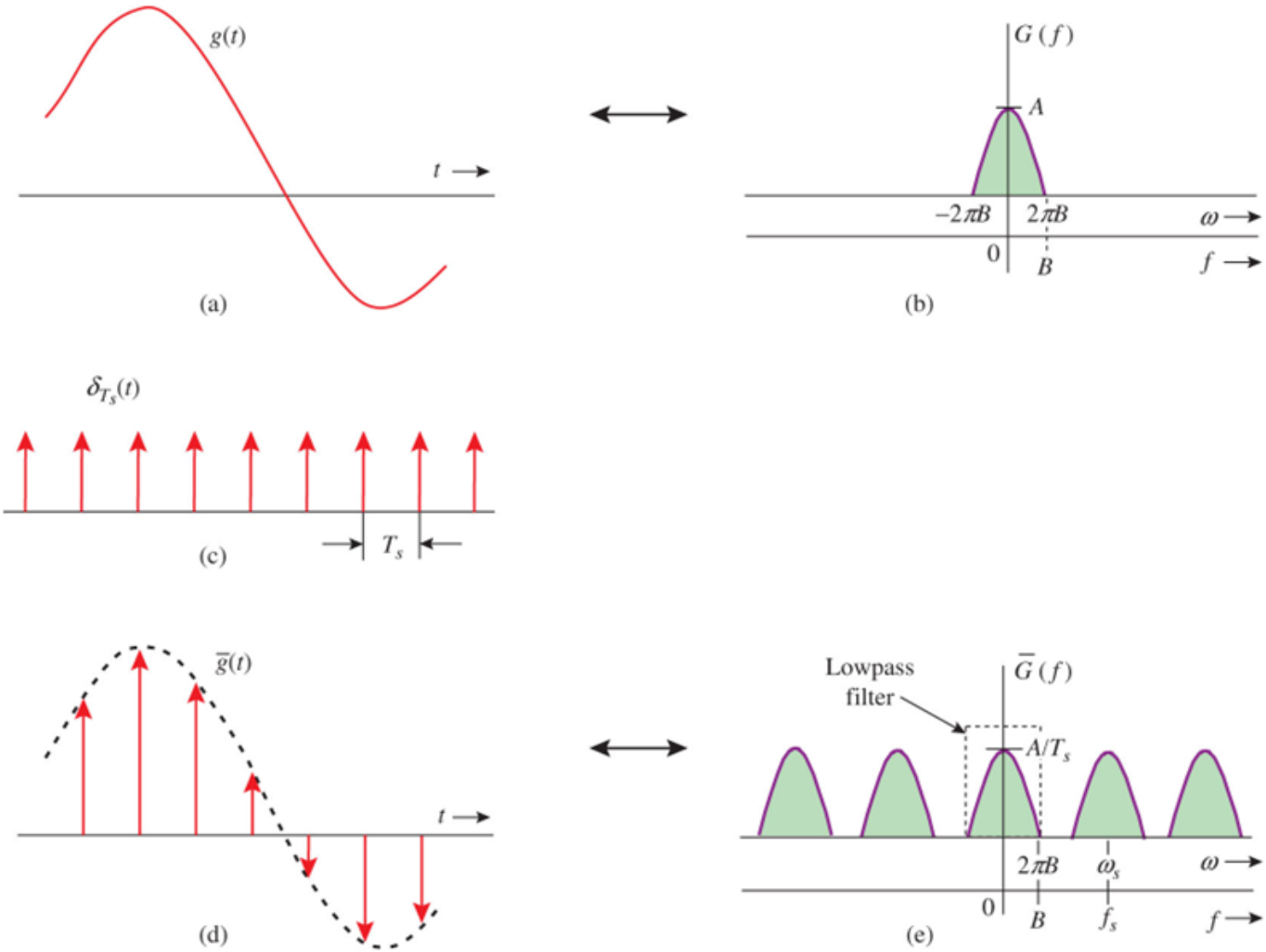
Lesson 18
Signal sampling and reconstruction



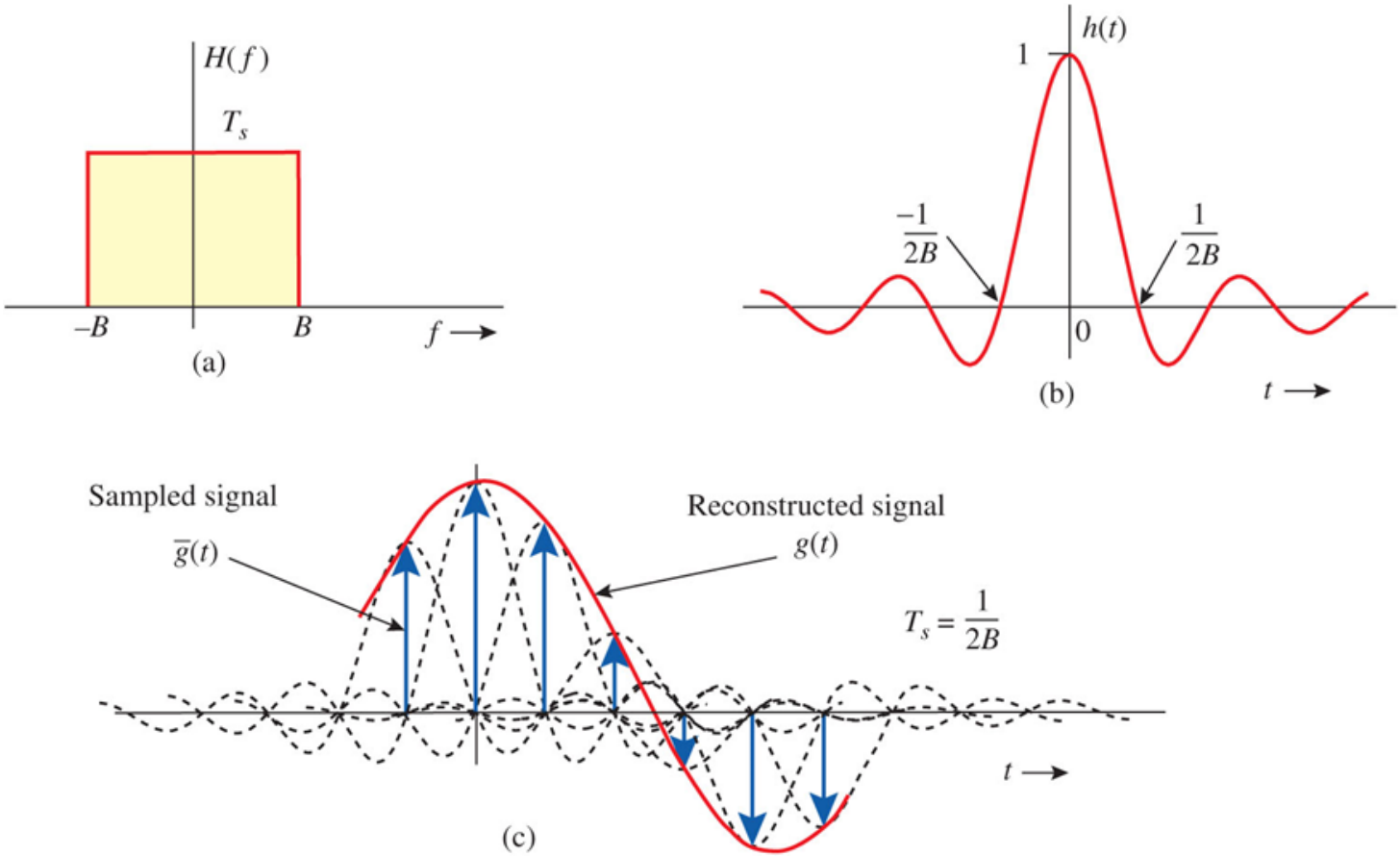
Schedule and Admin

- Schedule
- Admin
  - **HW4**. Assigned today. Due Lesson 21 (Tue, 6 Oct) to Gradescope.
  - **Lab 4**. Due Lesson 19 (Wed, 30 Sep) to Gradescope.
  - Everything from today through Lesson 32 is GR2 material (Chapters 5-8). GR2 is Lesson 35 (Tue, 17 Nov).

SAMPLING

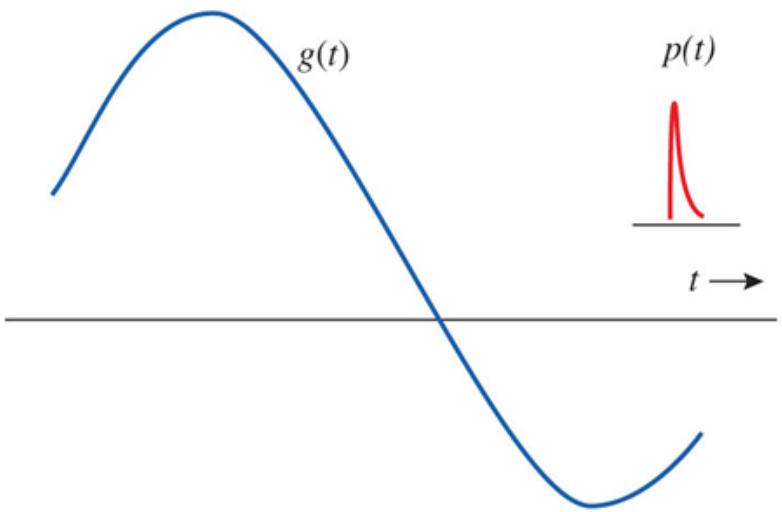

g(t)
t
(a)
G(f)
A
−2πB
2πB
ω
0
B
f
(b)
δ_Ts(t)
T_s
(c)
ḡ(t)
t
(d)
Lowpass filter
Ḡ(f)
A/T_s
2πB
ω_s
ω
0
B
f_s
f
(e)

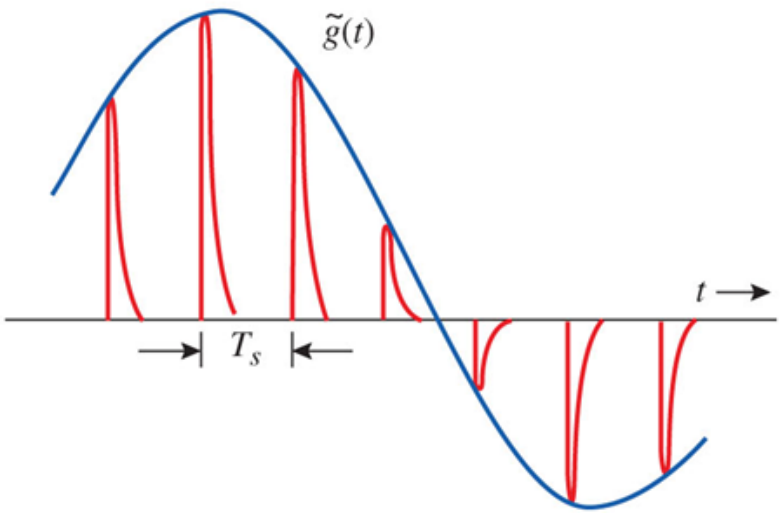
Ideal Reconstruction



Nonideal Reconstruction (Interpolation)

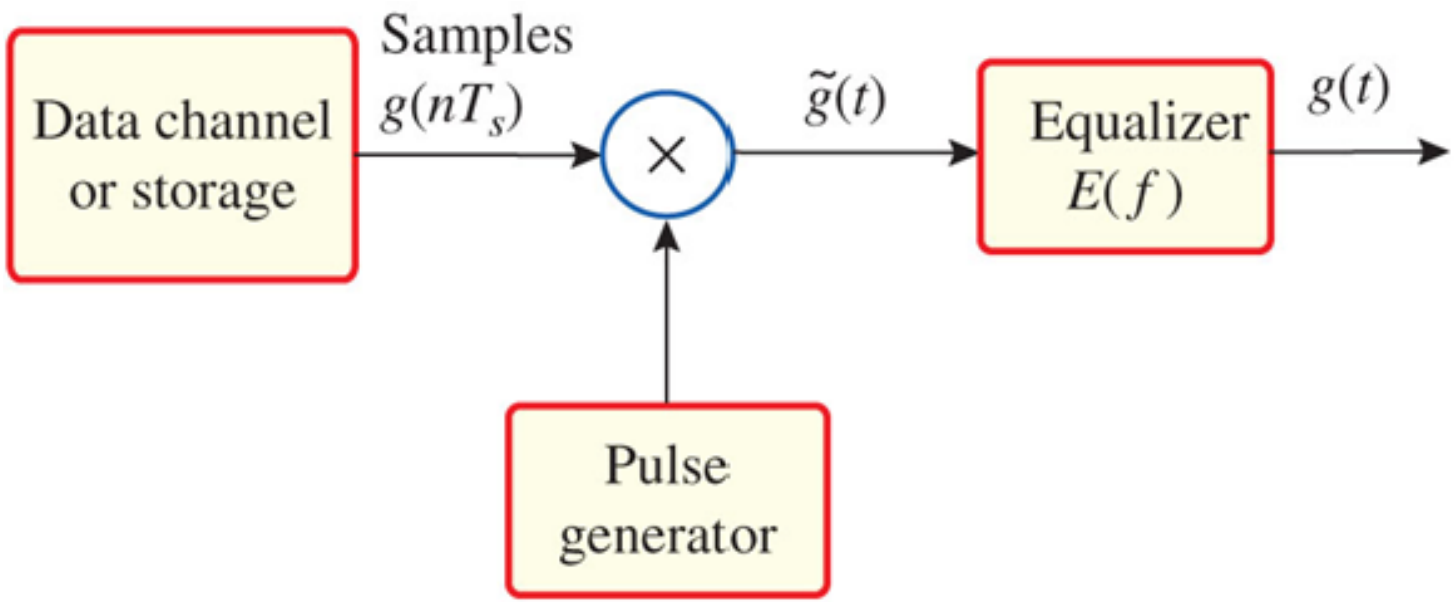
- Interpolation pulse $p(t)$
- $\tilde{g}(t) = p(t) * \bar{g}(t)$
- $\tilde{G}(f) = P(f)\bar{G}(f) = P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$





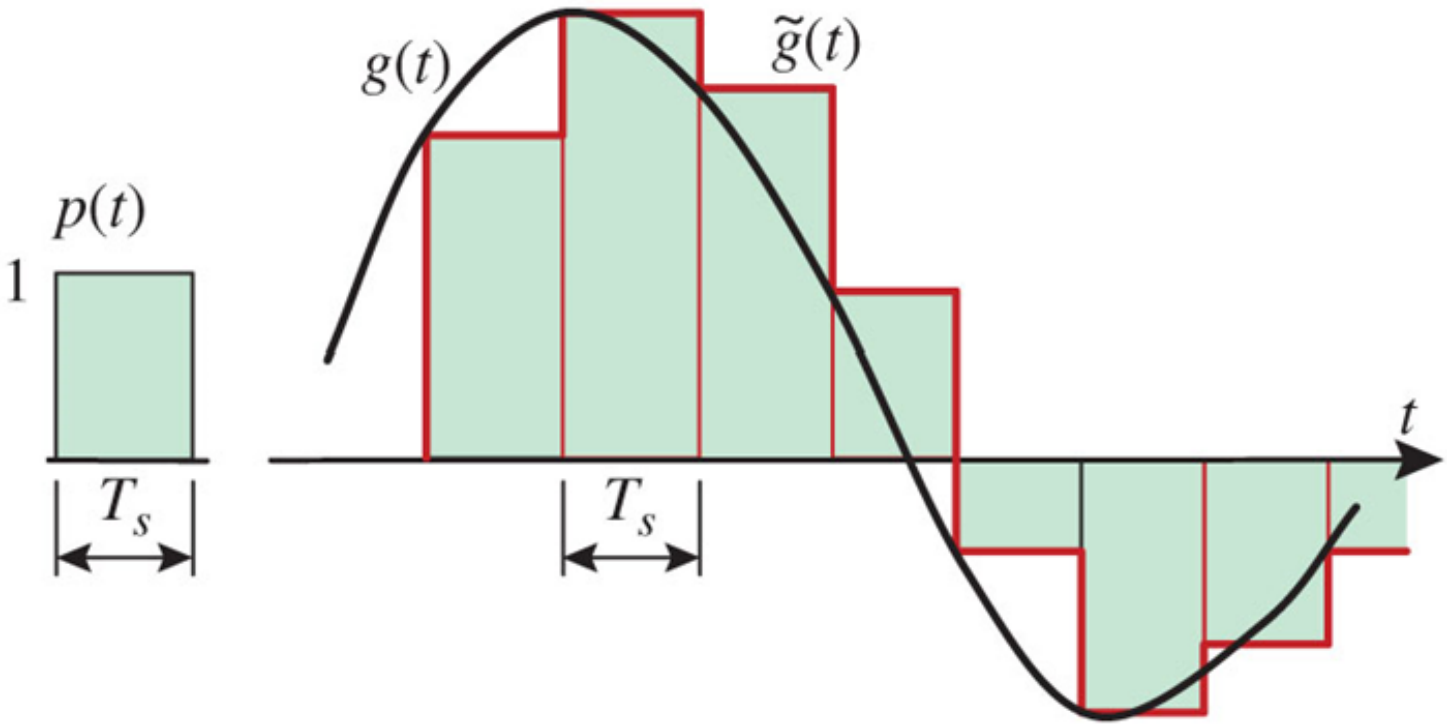
EQUALIZERS

- $P(f)$ filters sampled signal - need to reverse filtering to obtain original
- Use equalizers!



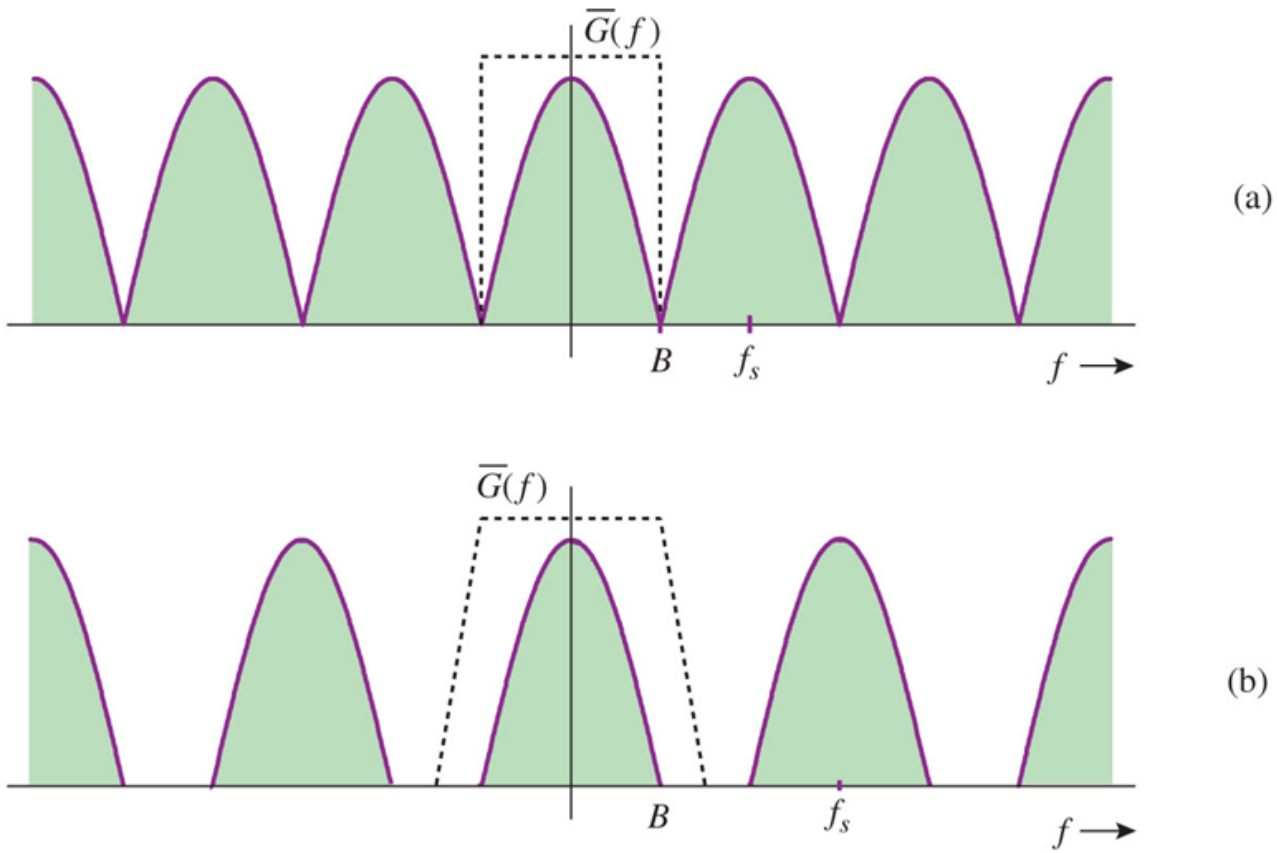
Interpolator Example - Flat Top Pulses

- Also known as a zero-order-hold pulse
- Samples signal and holds that value for T_s

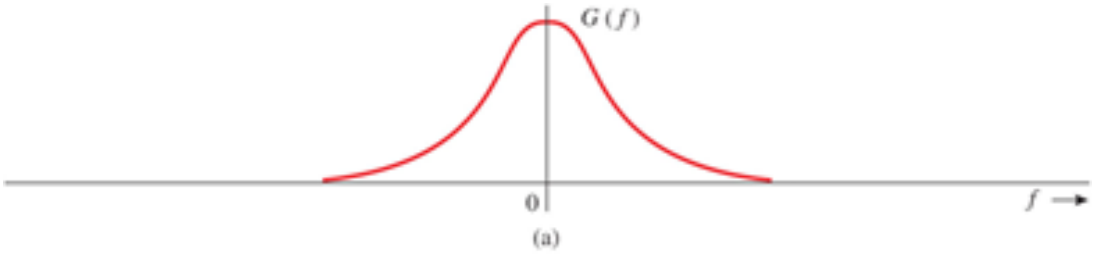


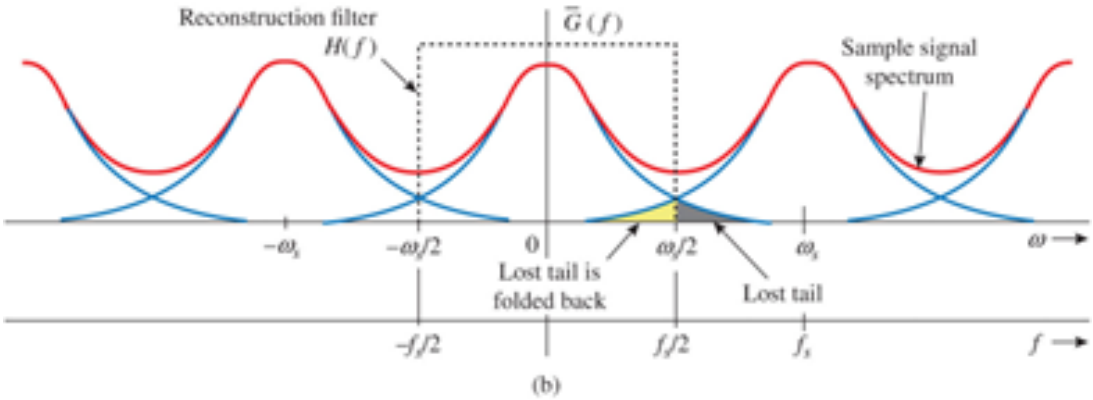
Realizable Reconstruction Filters

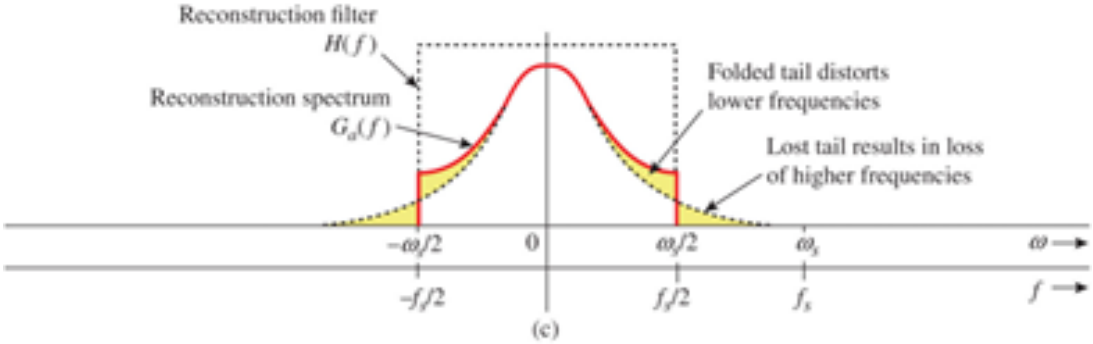
- Can't use ideal LPFs
- Need space between samples in spectra



ALIASING


G(f)
0
f
(a)


Reconstruction filter
H(f)
Ḡ(f)
Sample signal spectrum
−ω_s
−ω_s/2
0
ω_s/2
ω_s
ω
Lost tail is folded back
Lost tail
−f_s/2
f_s/2
f_s
f
(b)


Reconstruction filter
H(f)
Reconstruction spectrum
G_a(f)
Folded tail distorts lower frequencies
Lost tail results in loss of higher frequencies
−ω_s/2
0
ω_s/2
ω_s
ω
−f_s/2
f_s/2
f_s
f
(c)


Antialiasing filter
g(t)
H_aa(f)
g_aa(t)
Sampler
ḡ_aa(t)
δ_T(t)

Example Problem - 5.1-7